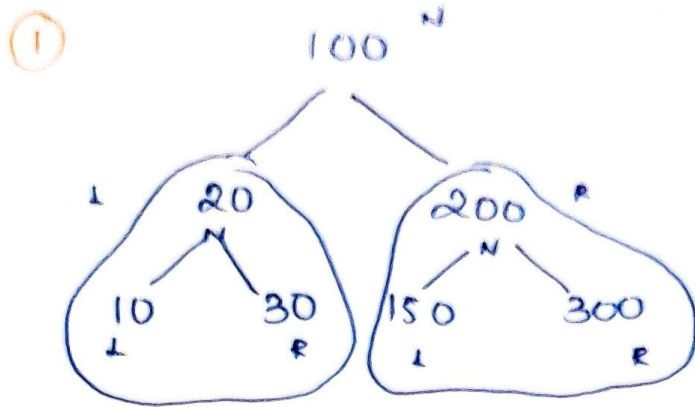


14/8/2026

Kiruthika B
192511232

CO3A13: Tree traversal challenge



Inorder: LNR $\Rightarrow$ $\frac{10 \quad 20 \quad 30}{L} \quad \frac{100}{N}$

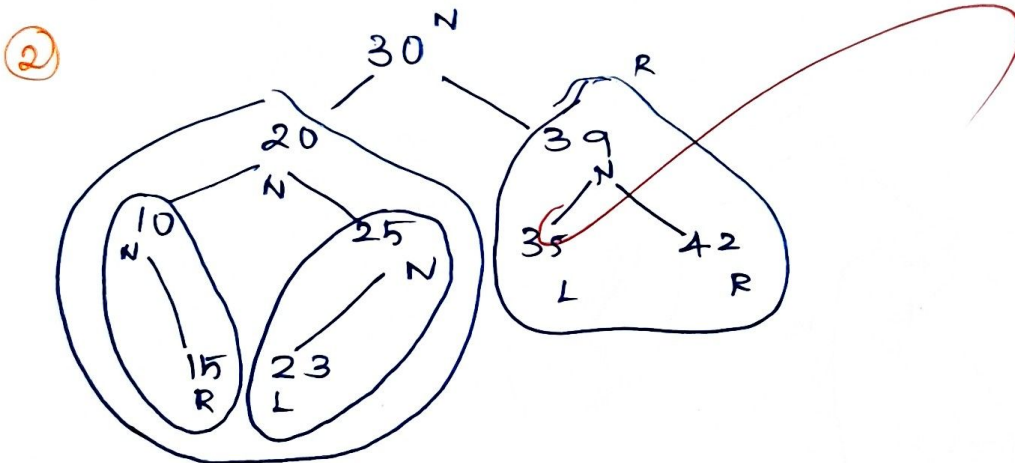
$\frac{150 \quad 200 \quad 300}{R}$

Preorder: NHR $\Rightarrow$ $\frac{100}{N} \quad \frac{20 \quad 10 \quad 30}{L}$

$\frac{200 \quad 150 \quad 300}{R}$

Postorder: LRN $\Rightarrow$ $\frac{10 \quad 30 \quad 20}{L} \quad \frac{100}{N}$

$\frac{150 \quad 300 \quad 200}{R} \quad \frac{100}{N}$



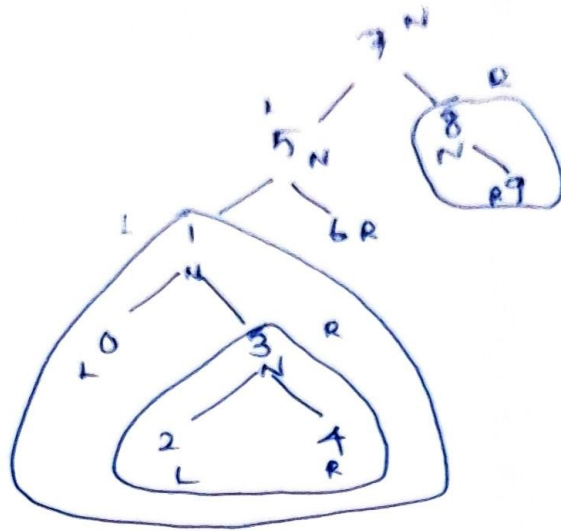
Inorder: LNR $\Rightarrow$ $\frac{10 \quad 15 \quad 20 \quad 23 \quad 25}{L} \quad \frac{30}{N} \quad \frac{35 \quad 39 \quad 42}{R}$

Preorder: NLR $\Rightarrow$ $\frac{30}{N} \quad \frac{20 \quad 10 \quad 15 \quad 25 \quad 23}{L} \quad \frac{39 \quad 35 \quad 42}{R}$

Postorder: LRN $\Rightarrow$ $\frac{15 \quad 10 \quad 23 \quad 25 \quad 20}{L} \quad \frac{35 \quad 42 \quad 39}{R} \quad \frac{30}{N}$

$\frac{35 \quad 42 \quad 39}{R} \quad \frac{30}{N}$

③ 7 5 1 8 3 6 0 9 4 2 → Binary Search

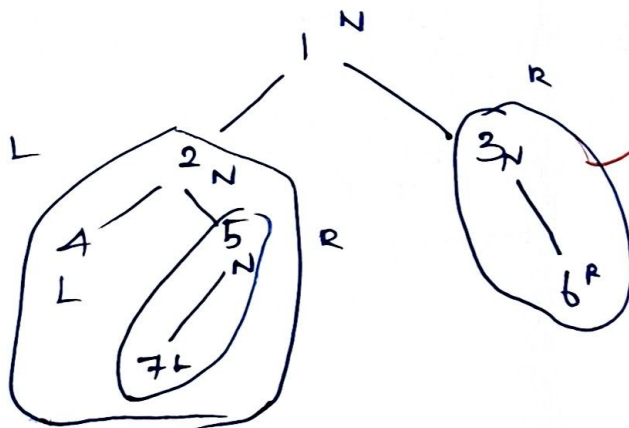


Inorder: LNR ⇒ $\frac{0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6}{L} \quad \frac{7 \ 8 \ 9}{N \ R}$

Preorder: NLR ⇒ $\frac{7}{N} \quad \frac{5 \ 1 \ 0 \ 3 \ 2 \ 4 \ 6}{L} \quad \frac{8 \ 9}{R}$

Postorder: LRN ⇒ $\frac{0 \ 2 \ 4 \ 3 \ 1 \ 6 \ 5}{L} \quad \frac{9 \ 8}{R} \quad \frac{7}{N}$

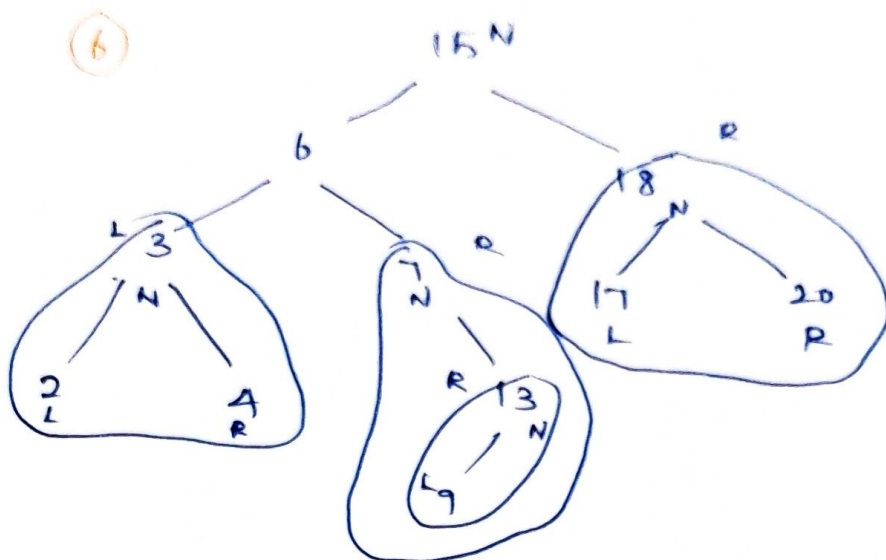
⑤



Inorder: LNR $\Rightarrow$ $\frac{7}{L} \frac{1}{N} \frac{2}{R} \frac{5}{R} \frac{1}{N} \frac{36}{R}$

Preorder: NLR $\Rightarrow$ $\frac{1}{N} \frac{7}{L} \frac{2}{R} \frac{5}{R} \frac{36}{R}$

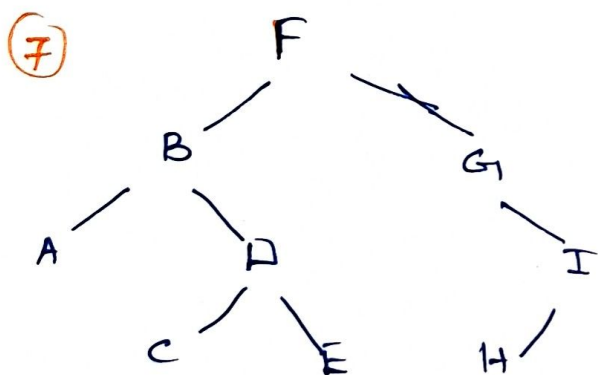
Postorder: LRN $\Rightarrow$ $\frac{7}{L} \frac{1}{N} \frac{5}{R} \frac{2}{R} \frac{36}{R} \frac{1}{N}$



Inorder: LNR $\Rightarrow$ $\frac{2}{L} \frac{3}{L} \frac{4}{L} \frac{6}{L} \frac{7}{L} \frac{9}{L} \frac{13}{L} \frac{15}{N} \frac{17}{R} \frac{18}{R}$

Preorder: NLR $\Rightarrow$ $\frac{15}{N} \frac{6}{L} \frac{3}{L} \frac{2}{L} \frac{4}{L} \frac{7}{L} \frac{13}{L} \frac{9}{L} \frac{18}{R} \frac{17}{R}$

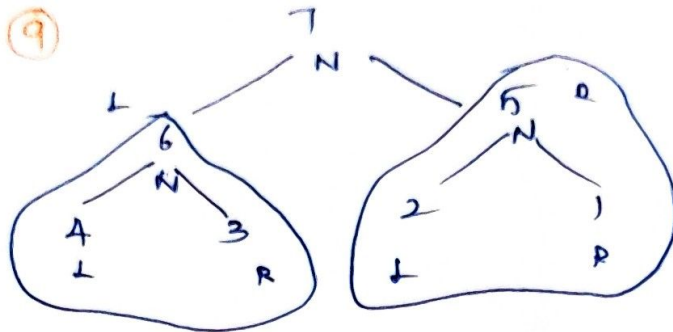
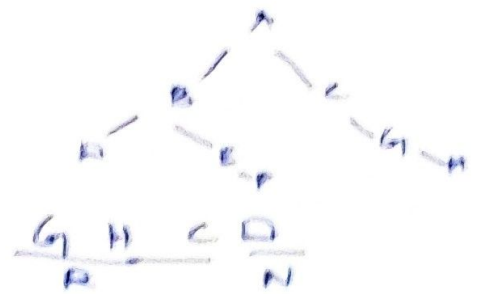
Post order: LRN $\Rightarrow$ $\frac{2}{L} \frac{4}{L} \frac{3}{L} \frac{9}{L} \frac{13}{L} \frac{7}{L} \frac{6}{L} \frac{17}{R} \frac{20}{R} \frac{18}{R}$



Inorder: LNR $\Rightarrow$ $\frac{A}{L} \frac{B}{L} \frac{C}{L} \frac{D}{L} \frac{E}{L} \frac{F}{N} \frac{G}{R} \frac{H}{R} \frac{I}{R}$

⑧ Inorder - AB E F A G H C
 Preorder - A B D E F C G H

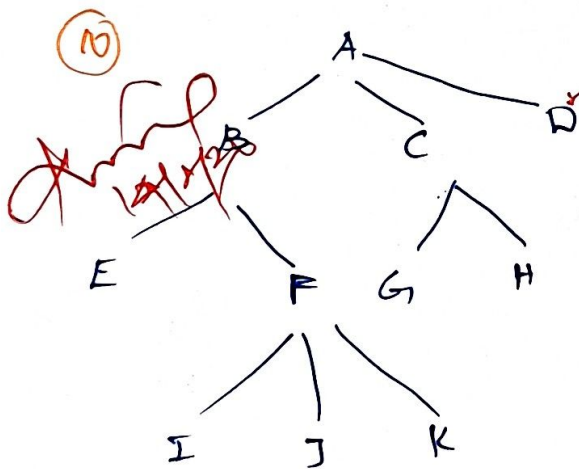
Postorder: LRN $\Rightarrow$ $\frac{D F E B}{L}$



Inorder: - LNR $\Rightarrow$ $\frac{4 \ 6 \ 3}{L}$ $\frac{7}{N}$ $\frac{2 \ 5 \ 1}{R}$

Preorder: NLR $\Rightarrow$ $\frac{7}{N}$ $\frac{6 \ 4 \ 3}{L}$ $\frac{5 \ 2 \ 1}{R}$

Postorder: LRN $\Rightarrow$ $\frac{4 \ 3 \ 6}{L}$ $\frac{2 \ 1 \ 5}{R}$ $\frac{7}{N}$



Inorder: LNR $\Rightarrow$ Not defined

Preorder: NLR $\Rightarrow$ A B E F I J K C G H D

Postorder: LRN $\Rightarrow$ E I J K F B G H C D A